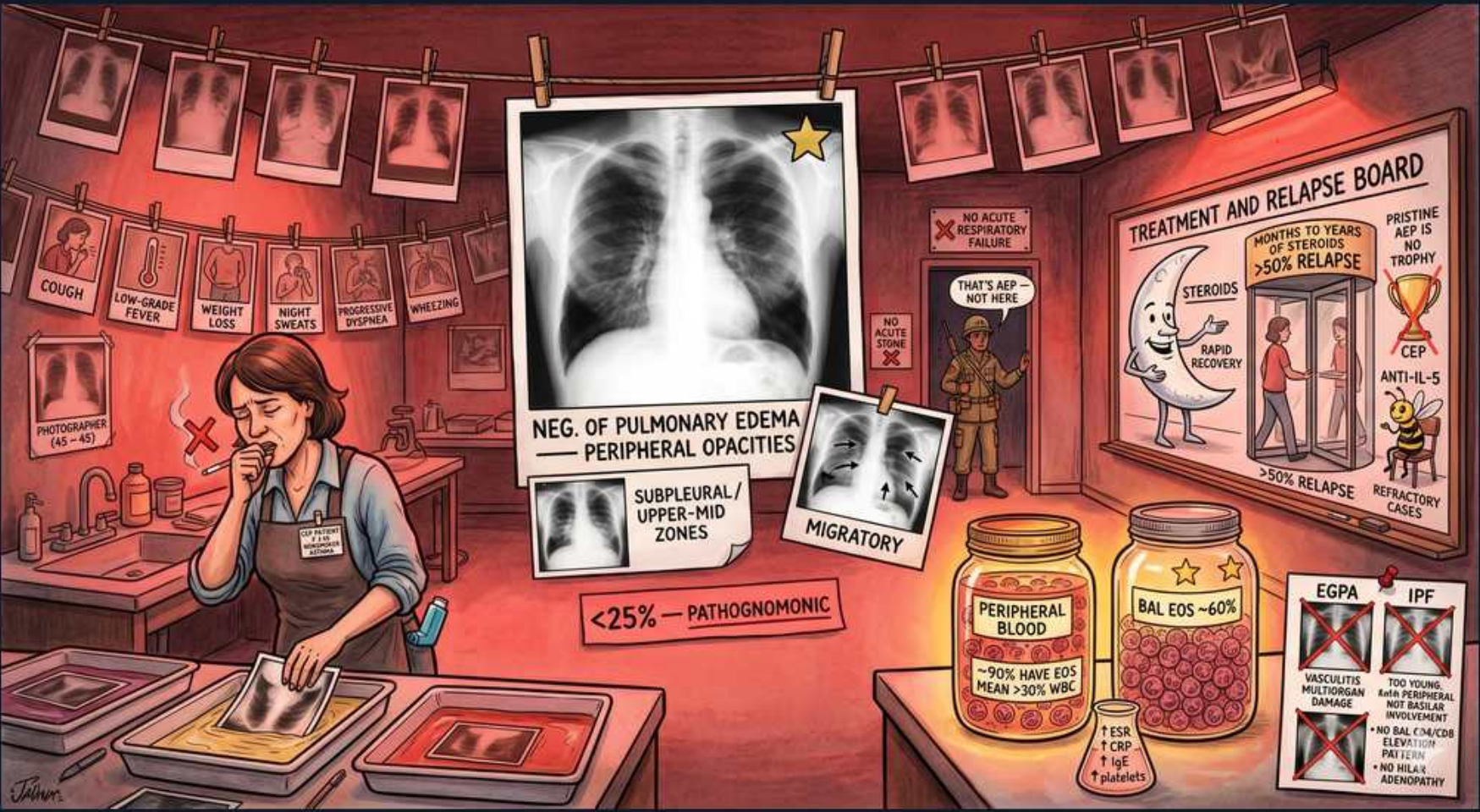


Eosinophilic Pneumonia — Chronic (CEP)



1. Neg. of pulmonary edema

WHAT YOU SEE

Central CXR labeled NEG. OF PULMONARY EDEMA, PERIPHERAL OPACITIES

WHAT IT ENCODES

CEP's classic radiographic sign is the 'photographic negative of pulmonary edema': peripheral (outer-zone) consolidations/opacities that SPARE the central perihilar region — the mirror image of cardiogenic pulmonary edema, which is central/perihilar ('bat-wing'). This pattern in a subacute cougher with eosinophilia is nearly diagnostic. Mnemonic: 'Chronic Eosinophilic = edges, not center.'

BOARD TRAP

WRONG answer: reading central perihilar 'bat-wing' opacities and calling it CEP. CEP is the photographic NEGATIVE — peripheral opacities sparing the center. Central bat-wing edema = cardiogenic pulmonary edema, not CEP.

2. Subpleural upper-mid zones

WHAT YOU SEE

Tag SUBPLEURAL/UPPER-MID ZONES

WHAT IT ENCODES

CEP opacities are subpleural and upper/mid-lung predominant — the classic CEP distribution. This contrasts with IPF and most fibrotic ILDs, which are basal/subpleural and lower-zone predominant. Upper-zone peripheral consolidation in an asthmatic woman with eosinophilia points to CEP.

BOARD TRAP

WRONG answer: attributing upper-lobe peripheral consolidations to IPF or postprimary TB. IPF is lower-zone fibrotic and steroid-unresponsive; CEP is upper/mid peripheral, eosinophilic, and melts away with steroids.

3. Migratory opacities

WHAT YOU SEE

Small film tagged MIGRATORY

WHAT IT ENCODES

CEP infiltrates can be migratory, shifting location over serial films — overlapping with the migratory infiltrates of Löffler/simple pulmonary eosinophilia and organizing pneumonia. In CEP they are more persistent and peripheral, and accompanied by blood eosinophilia and systemic symptoms.

BOARD TRAP

WRONG answer: calling transient, rapidly migrating infiltrates with minimal symptoms CEP. Fleeting migratory infiltrates that clear on their own suggest Löffler (simple pulmonary eosinophilia); CEP infiltrates are more fixed, peripheral, symptomatic, and steroid-requiring.

4. Middle-aged woman

WHAT YOU SEE

A middle-aged woman in a darkroom developing chest X-ray films, badge reading '45–45' and a cigarette — she is the classic CEP patient: 40s–50s, female, often asthmatic.

WHAT IT ENCODES

Classic CEP patient: a middle-aged woman (40s–50s), frequently with asthma/atopy. Female predominance (~2:1) and a background of asthma are high-yield demographic clues that separate CEP from AEP (young male, new smoker, no asthma).

BOARD TRAP

WRONG answer: expecting a young male new-smoker — that is the AEP profile. CEP favors middle-aged atopic/asthmatic women with a subacute course.

5. Asthma / atopy history

WHAT YOU SEE

Left-wall strip of small tags including a WHEEZING tag with an inhaler icon — the wheeze/inhaler flags the preceding asthma/atopy background most CEP patients carry.

WHAT IT ENCODES

Most CEP patients (about half or more) have a preceding history of asthma or atopy; asthma may worsen as CEP develops. This atopic background is a key distinguishing feature from AEP and aligns CEP on the eosinophilic-asthma/allergic spectrum that also includes EGPA and ABPA.

BOARD TRAP

WRONG answer: using asthma history to diagnose EGPA instead of CEP. Asthma occurs in both, but EGPA adds systemic vasculitis (neuropathy, sinus, cardiac, ANCA); isolated lung disease with peripheral eosinophilia and the photographic-negative pattern is CEP.

6. Subacute symptoms

WHAT YOU SEE

Tags COUGH, LOW-GRADE FEVER, WEIGHT LOSS, NIGHT SWEATS

WHAT IT ENCODES

CEP is subacute-to-chronic: weeks to months of cough, dyspnea, wheeze, low-grade fever, weight loss, and night sweats. The 'B-symptom' constitutional picture plus weight loss often triggers a TB or malignancy workup before eosinophilia and the imaging pattern reveal CEP. Duration (weeks-months) is the key contrast with AEP (<1 month, acute).

BOARD TRAP

WRONG answer: working the chronic-cough/weight-loss/night-sweats picture as TB or lymphoma and missing CEP. The peripheral blood eosinophilia plus peripheral upper-zone (photographic-negative) opacities reorient the diagnosis to CEP.

7. Peripheral blood eos

WHAT YOU SEE

Jar PERIPHERAL BLOOD >90% have eos, mean >30%

WHAT IT ENCODES

CEP has peripheral blood eosinophilia in >90% of patients (peripheral eosinophils commonly making up a large fraction of the differential, mean >30%) — a defining laboratory feature and a major contrast with AEP, where early peripheral eosinophilia is absent. Blood eosinophilia plus the imaging pattern often makes the diagnosis without biopsy.

BOARD TRAP

WRONG answer: thinking the absence of peripheral eosinophilia is typical (that is AEP). In CEP, peripheral blood eosinophilia is present in >90% and is one of the most reliable clues — a normal eosinophil count argues against CEP.

8. BAL eos >40%

WHAT YOU SEE

Jar BAL EOS >40% (>60% star)

WHAT IT ENCODES

BAL in CEP shows marked eosinophilia (commonly >40%, sometimes >60%) — even higher than the >25% threshold used for AEP. With the typical clinical/imaging picture, this degree of BAL eosinophilia confirms CEP and obviates lung biopsy.

BOARD TRAP

WRONG answer: requiring a surgical lung biopsy to diagnose CEP. The combination of peripheral eosinophilia, photographic-negative imaging, and high BAL eosinophilia is sufficient — biopsy is usually unnecessary.

9. <25% pathognomonic

WHAT YOU SEE

Banner <25% — PATHOGNOMONIC

WHAT IT ENCODES

A high BAL eosinophil percentage in the correct clinical setting is essentially pathognomonic for eosinophilic pneumonia. The threshold matters: >25% defines the eosinophilic pneumonias, and the very high percentages seen in CEP (often >40%) virtually exclude other ILDs in the differential.

BOARD TRAP

WRONG answer: dismissing eosinophilic pneumonia because BAL eosinophils are 'only' mildly elevated. Mild BAL eosinophilia is nonspecific; it is the >25% (and in CEP often >40%) threshold that is diagnostically powerful.

10. High relapse on taper

WHAT YOU SEE

Board TREATMENT AND RELAPSE, >50% RELAPSE

WHAT IT ENCODES

CEP relapses in >50% of patients as steroids are tapered or stopped, so most patients need a prolonged course (often many months) and some need long-term low-dose maintenance. This high relapse rate is the single best feature distinguishing CEP from AEP, which does not recur.

BOARD TRAP

WRONG answer: stopping steroids after a couple of weeks expecting a one-and-done cure (that is AEP behavior). CEP relapses in >50% on taper — premature discontinuation predictably brings the disease back.

11. Steroids — rapid recovery

WHAT YOU SEE

Moon STEROIDS, RAPID RECOVERY

WHAT IT ENCODES

CEP responds dramatically and rapidly to oral corticosteroids — prednisone ~0.5 mg/kg/day (roughly 40-60 mg/day) with symptom and imaging improvement within days, sometimes hours. The catch is the slow taper over months because of the high relapse risk; rapid initial response does not mean short treatment.

BOARD TRAP

WRONG answer: equating rapid response with a short course. The brisk response is real, but unlike AEP the taper must be prolonged (months) to prevent relapse.

12. Anti-IL-5 sparing

WHAT YOU SEE

Bee icon ANTI-IL-5

WHAT IT ENCODES

For relapsing or steroid-dependent CEP, anti-IL-5/IL-5-receptor biologics (mepolizumab, reslizumab, benralizumab) act as steroid-sparing agents — IL-5 is the key cytokine driving eosinophil maturation and survival. This mirrors their use in severe eosinophilic asthma and EGPA.

BOARD TRAP

WRONG answer: reaching for cyclophosphamide or a broad immunosuppressant for steroid-dependent CEP. CEP is not a vasculitis; the targeted steroid-sparing approach is anti-IL-5 biologic therapy, not cytotoxic immunosuppression.

13. No acute resp failure

WHAT YOU SEE

Crossed tag NO ACUTE RESP FAILURE

WHAT IT ENCODES

Unlike AEP, CEP rarely causes acute hypoxemic respiratory failure or an ARDS picture — its course is subacute/insidious, and patients are usually not ventilator-dependent at presentation. The slow tempo plus peripheral eosinophilia frames CEP as a chronic, steroid-managed disease rather than an ICU emergency.

BOARD TRAP

WRONG answer: expecting fulminant respiratory failure to confirm CEP. Acute respiratory failure/ARDS picture points to AEP; CEP presents subacutely and seldom needs ventilation.

14. Elevated ESR/CRP/IgE

WHAT YOU SEE

Labels ESR, CRP, IgE, platelets up

WHAT IT ENCODES

Supportive (nonspecific) labs in CEP: elevated ESR and CRP, elevated serum IgE (reflecting the atopic/allergic background), and thrombocytosis, in addition to the peripheral eosinophilia. These reinforce the inflammatory eosinophilic picture but are not specific on their own.

BOARD TRAP

WRONG answer: using elevated inflammatory markers to call this a generic infection or to exclude an eosinophilic process. In CEP, raised ESR/CRP/IgE are expected and supportive; the eosinophilia and imaging pattern are what specify the diagnosis.

15. EGPA mimic

WHAT YOU SEE

Panel EGPA — vasculitis, multisystem, ANCA

WHAT IT ENCODES

Differential: EGPA (eosinophilic granulomatosis with polyangiitis, formerly Churg-Strauss) — an eosinophilic ANCA-associated vasculitis with asthma plus multisystem features (mononeuritis multiplex, sinusitis, cardiac, GI, skin purpura, glomerulonephritis); p-ANCA/MPO is positive in ~40-60%. CEP overlaps (asthma + eosinophilia + lung infiltrates) but is lung-limited without vasculitis.

BOARD TRAP

WRONG answer: labeling lung-limited eosinophilic pneumonia as EGPA. The discriminator is systemic vasculitis: neuropathy, sinus disease, cardiac/renal involvement, or ANCA positivity indicate EGPA; their absence with isolated peripheral lung disease indicates CEP.

16. IPF mimic

WHAT YOU SEE

Panel IPF — too young, no peripheral eos

WHAT IT ENCODES

Differential: idiopathic pulmonary fibrosis (IPF) — but IPF affects older patients (typically >60), shows lower-zone/basal subpleural reticulation with honeycombing (UIP pattern), has NO eosinophilia, and is steroid-UNRESPONSIVE (treated with antifibrotics pirfenidone/nintedanib). CEP is younger, upper/mid peripheral, eosinophilic, and steroid-responsive.

BOARD TRAP

WRONG answer: starting antifibrotics (pirfenidone/nintedanib) or withholding steroids because you assumed IPF. IPF is basal/fibrotic, non-eosinophilic, and steroid-unresponsive; CEP's peripheral upper-lobe eosinophilic infiltrates resolve with corticosteroids — completely different management.
